Regression and Predictive Analytics

Stat 3502, Fall 2024

Regression models

- A correctly specified regression model contains all of the relevant predictors
- An underspecified regression model is missing one or more important predictor variables
- An overspecified regression model has one or more redundant predictor variables. The redundant predictors are generally correlated with the "true" predictors
- A regression model may contain one or more extraneous variables. An extraneous variable is not related to the response or any of the other predictors.

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- Model is more complicated and harder to understand than necessary
- MSE ($\hat{\sigma}^2$) will be bigger than the correctly specified model
- Confidence intervals will be wider
- Hypothesis tests will have smaller power (harder to reject the null when H_0 is false)

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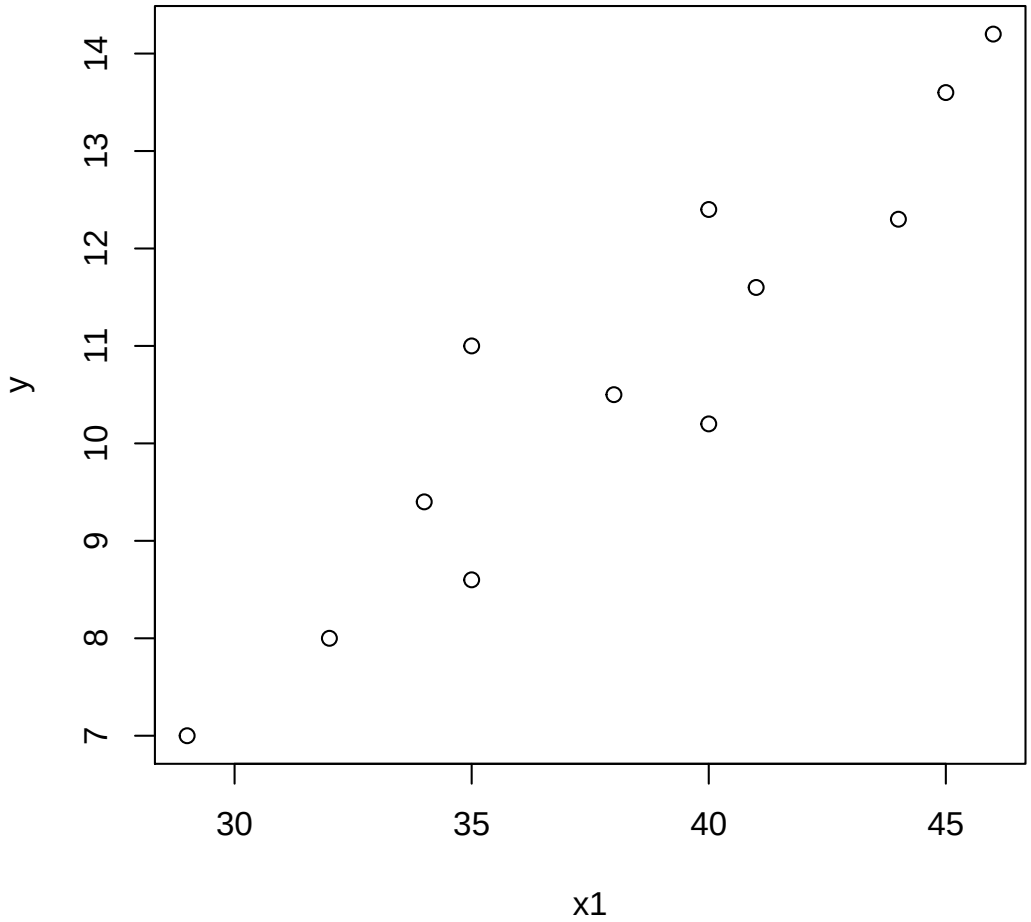
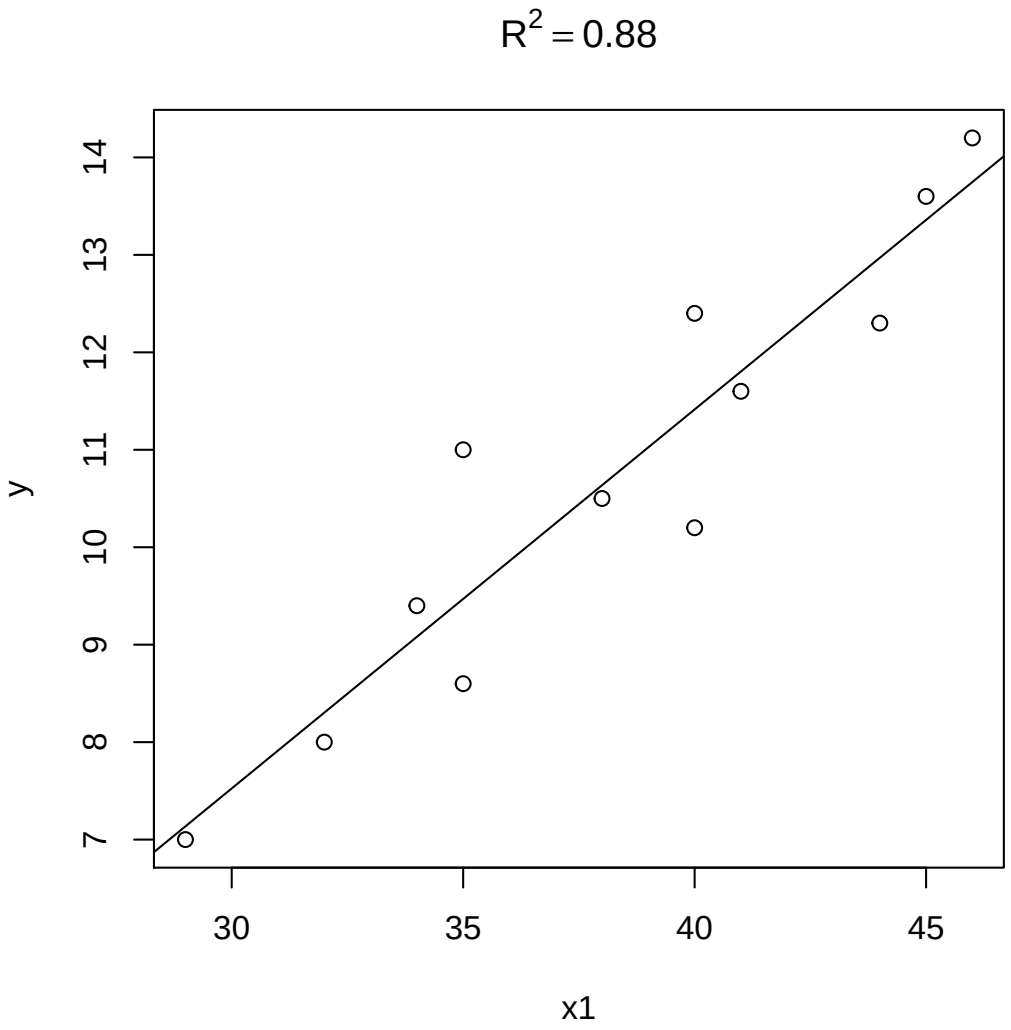
Underspecified models

Refer to under.txt and under.r.



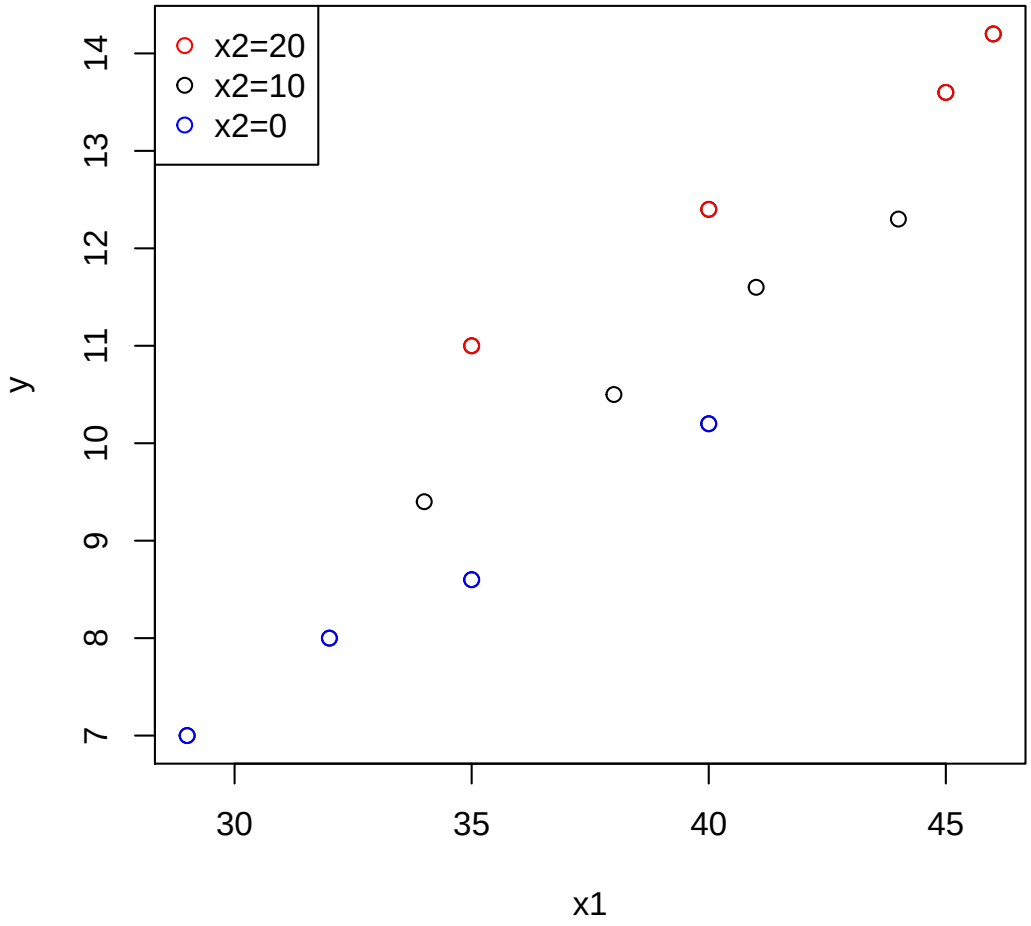
Underspecified models

Fit a model between y and $x1$



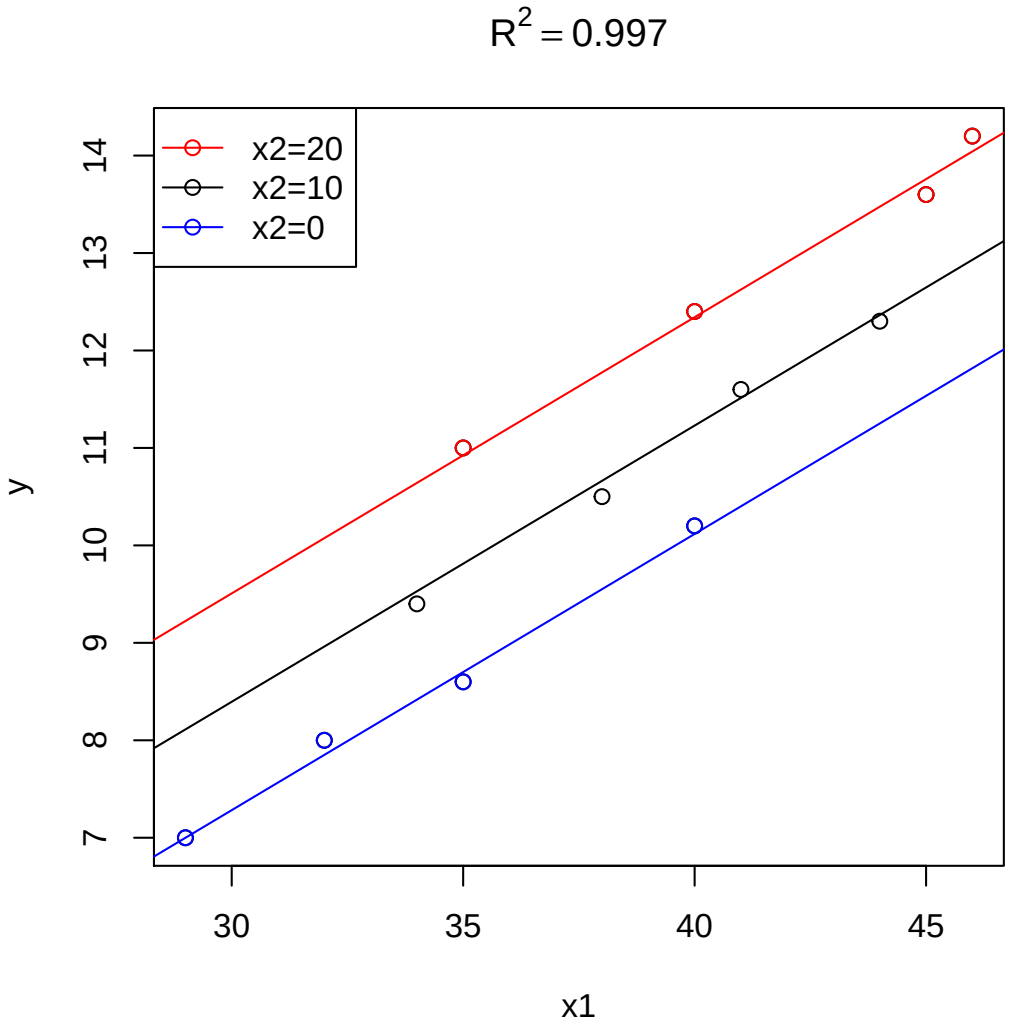
Underspecified models

There is actually $x2$



Underspecified models

Fit a model with both $x1$ and $x2$



Underspecified models

- Estimated regression coefficients ($\hat{\beta}$'s) are biased
- R^2 will be smaller than the correctly specified model
- MSE will be bigger than the correctly specified model

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All possible regressions

- Suppose there is one predictor, there are two possible models: intercept only model and simple linear regression
- How many possible models are there if there are two predictors x_1 and x_2?
  intercept only, SLR with x_1 or x_2, MLR with both x_1 and x_2
- In general, 2^k possible models if there are k predictors

Cement data

Refer to cement.txt

y: heat evolved in calories during hardening of cement (per gram)

x_1: percentage of tricalcium aluminate

x_2: percentage of tricalcium silicate

x_3: percentage of tetracalcium alumino ferrite

x_4: percentage of dicalcium silicate

Cement data

Variables in Model	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	$\hat{\beta}_4$
x_1	81.479	1.869			
x_2	57.424		0.789		
x_3	110.203			–1.256	
x_4	117.568				–0.738
x_1x_2	52.577	1.468	0.662		
x_1x_3	72.349	2.312		0.494	
x_1x_4	103.097	1.440			–0.614
x_2x_3	72.075		0.731	–1.008	
x_2x_4	94.160		0.311		–0.457
x_3x_4	131.282			–1.200	–0.724
$x_1x_2x_3$	48.194	1.696	0.657	0.250	
$x_1x_2x_4$	71.648	1.452	0.416		–0.237
$x_2x_3x_4$	203.642		–0.923	–1.448	–1.557
$x_1x_3x_4$	111.684	1.052		–0.410	–0.643
$x_1x_2x_3x_4$	62.405	1.551	0.510	0.102	–0.144

Cement data

The signs of $\hat{\beta}_2$ and $\hat{\beta}_3$ change across different models. Why?

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There is multicollinearity

	x_1	x_2	x_3	x_4	y
x_1	1.0				
x_2	0.229	1.0			
x_3	–0.824	–0.139	1.0		
x_4	–0.245	–0.973	0.030	1.0	
y	0.731	0.816	–0.535	–0.821	1.0

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Big correlations between x_1 and x_3, as well as between x_2 and x_4

Criteria for evaluating regression models

Recall k is the number of predictors and $p = k + 1$

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Bigger is better:

- Coefficient of multiple determination R_p^2
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Smaller is better:

- Mean squared error MSE, or mean squared residual $MS_{\mathrm{Res}}(p)$
- Mallow's C_p criterion

Cement data

Number of Regressors in Model	p	Regressors in Model	$SS_{\text{Res}}(p)$	R_p^2	$R_{\text{Adj},p}^2$	$MS_{\text{Res}}(p)$	C_p
None	1	None	2715.7635	0	0	226.3136	442.92
1	2	x_1	1265.6867	0.53395	0.49158	115.0624	202.55
1	2	x_2	906.3363	0.66627	0.63593	82.3942	142.49
1	2	x_3	1939.4005	0.28587	0.22095	176.3092	315.16
1	2	x_4	883.8669	0.67459	0.64495	80.3515	138.73
2	3	x_1x_2	57.9045	0.97868	0.97441	5.7904	2.68
2	3	x_1x_3	1227.0721	0.54817	0.45780	122.7073	198.10
2	3	x_1x_4	74.7621	0.97247	0.96697	7.4762	5.50
2	3	x_2x_3	415.4427	0.84703	0.81644	41.5443	62.44
2	3	x_2x_4	868.8801	0.68006	0.61607	86.8880	138.23
2	3	x_3x_4	175.7380	0.93529	0.92235	17.5738	22.37
3	4	$x_1x_2x_3$	48.1106	0.98228	0.97638	5.3456	3.04
3	4	$x_1x_2x_4$	47.9727	0.98234	0.97645	5.3303	3.02
3	4	$x_1x_3x_4$	50.8361	0.98128	0.97504	5.6485	3.50
3	4	$x_2x_3x_4$	73.8145	0.97282	0.96376	8.2017	7.34
4	5	$x_1x_2x_3x_4$	47.8636	0.98238	0.97356	5.9829	5.00

Cement data

Different criteria may lead to different results

- According to R_p^2 or $R_{\mathrm{Adj},p}^2$, the following models are almost equally good:
  x_1x_2, $x_1x_2x_3$, $x_1x_2x_4$, $x_1x_3x_4$ and $x_1x_2x_3x_4$

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- According to $MS_{\mathrm{Res}}(p)$, x_1x_2, $x_1x_2x_3$, and $x_1x_2x_4$ are the best
- According to Mallow's C_p, x_1x_2 is the best

Best subset selection

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- Identify all possible regression models from all possible combinations of predictors
- Identify the best one-predictor model, the best two-predictor model, etc.
  Here "best" means any one of the criteria discussed earlier
- Further evaluate and refine the handful of models identified in the previous step. Such as residual analysis, transformations, and adding interaction terms

Prediction sum of squares and $R^2_{Prediction}$

- Prediction sum of squares is defined as

$$\text{PRESS} = \sum_{i=1}^{n}(y_i - \hat{y}_{(i)})^2 = \sum_{i=1}^{n}\left(\frac{e_i}{1-h_{ii}}\right)^2$$

  Recall h_{ii} is the leverage, e_i is the residual, and $\hat{y}_{(i)}$ is the predicted response using the model fit after deleting the ith observation

- Predicted R^2 is defined as

$$R^2_{\text{Prediction}} = 1 - \frac{\text{PRESS}}{SST}$$

  Here $SST = \sum_{i=1}^{n}(y_i - \bar{y})^2$ is the total sum of squares

Cement data

Observation i	$\hat{y} = 52.58 + 1.468x_1 + 0.662x_2$ [a]			$\hat{y} = 71.65 + 1.452x_1 + 0.416x_2 - 0.237x_4$ [b]		
	e_i	h_{ii}	$[e_i/(1-h_{ii})]^2$	e_i	h_{ii}	$[e_i/(1-h_{ii})]^2$
1	−1.5740	0.25119	4.4184	0.0617	0.52058	0.0166
2	−1.0491	0.26189	2.0202	1.4327	0.27670	3.9235
3	−1.5147	0.11890	2.9553	−1.8910	0.13315	4.7588
4	−1.6585	0.24225	4.7905	−1.8016	0.24431	5.6837
5	−1.3925	0.08362	2.3091	0.2562	0.35733	0.1589
6	4.0475	0.11512	20.9221	3.8982	0.11737	19.5061
7	−1.3031	0.36180	4.1627	−1.4287	0.36341	5.0369
8	−2.0754	0.24119	7.4806	−3.0919	0.34522	22.2977
9	1.8245	0.17195	4.9404	1.2818	0.20881	2.6247
10	1.3625	0.55002	9.1683	0.3539	0.65244	1.0368
11	3.2643	0.18402	16.0037	2.0977	0.32105	9.5458
12	0.8628	0.19666	1.1535	1.0556	0.20040	1.7428
13	−2.8934	0.21420	13.5579	−2.2247	0.25923	9.0194
		PRESS x_1, x_2 = 93.8827			PRESS x_1, x_2, x_4 = 85.3516	

[a] $R^2_{\text{Prediction}} = 0.9654$, $\text{VIF}_1 = 1.05$, $\text{VIF}_2 = 1.06$.

[b] $R^2_{\text{Prediction}} = 0.9684$, $\text{VIF}_1 = 1.07$, $\text{VIF}_2 = 18.78$, $\text{VIF}_4 = 18.94$.

Cement data

Best subset selection

- Choose the model with the smallest PRESS or the largest $R^2_{\mathrm{Prediction}}$
- Calculation of PRESS can be time-consuming
- PRESS and $R^2_{\mathrm{Prediction}}$ can be used to compare a limited number of models

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For cement data with x_1x_2 model and $x_1x_2x_4$ model

- Two models have similar PRESS
- Two models have similar $R^2_{\mathrm{Prediction}}$
- We prefer x_1x_2 model, as $x_1x_2x_4$ model has multicollinearity (VIF$>$10)

Forward selection and backward elimination

Build our regression model from a set of candidate predictor variables by entering and removing predictors

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- Forward selection starts from the intercept only model, then adds one predictor at each step
- Backward elimination starts from all predictors in the model, then deletes one predictor at each step

Forward selection

- Needs a criterion to decide which predictor to add at each step. For example, the one with the smallest p-value in the t-test
- Needs a criterion to stop the iteration. For example, a threshold for the p-value in the t-test

Forward selection for cement data

Refer to cement_forward.r

Suppose threshold to enter the model is $\alpha = 0.15$.

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- Start from intercept only model
- Compare four SLR models: x_1 only, x_2 only, x_3 only, and x_4 only. According to p-values in the t-test, x_4 is added (with p-value $0 < \alpha$)

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- Compare three models: $x_4 x_1$, $x_4 x_2$ and $x_4 x_3$. According to p-values in the t-test, x_1 is added (with p-value $0 < \alpha$)

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- Compare four SLR models: x_1 only, x_2 only, x_3 only, and x_4 only. According to p-values in the t-test, x_4 is added (with p-value $0 < \alpha$)
- Compare three models: x_4x_1, x_4x_2 and x_4x_3. According to p-values in the t-test, x_1 is added (with p-value $0 < \alpha$)
- Compare two models: $x_4x_1x_2$ and $x_4x_1x_3$. According to p-values in the t-test, x_2 is added (with p-value $0.05 < \alpha$)

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- Compare three models: x_4x_1, x_4x_2 and x_4x_3. According to p-values in the t-test, x_1 is added (with p-value $0 < \alpha$)
- Compare two models: $x_4x_1x_2$ and $x_4x_1x_3$. According to p-values in the t-test, x_2 is added (with p-value $0.05 < \alpha$)
- Given current model $x_4x_1x_2$, decide whether x_3 should be added. From model $x_1x_2x_3x_4$, p-value for x_3 is 0.896 ($> \alpha$). Conclude $x_1x_2x_4$ is the final model

Backward elimination

- Needs a criterion to decide which predictor to delete at each step. For example, the one with the largest p-value in the t-test
- Needs a criterion to stop the iteration. For example, a threshold for the p-value in the t-test

Backward elimination for cement data

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Suppose threshold for p-value is $\alpha = 0.15$.

- Start from $x_1 x_2 x_3 x_4$
- x_3 has the largest p-value according to t-test, x_3 is deleted (with p-value 0.896$> \alpha$)

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Suppose threshold for p-value is $\alpha = 0.15$.

- Start from $x_1 x_2 x_3 x_4$
- x_3 has the largest p-value according to t-test, x_3 is deleted (with p-value 0.896$> \alpha$)
- Consider t-tests from the $x_1 x_2 x_4$ model. x_4 has the largest p-value. x_4 is deleted (with p-value 0.205$> \alpha$)

Backward elimination for cement data

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Suppose threshold for p-value is $\alpha = 0.15$.

- Start from $x_1 x_2 x_3 x_4$
- x_3 has the largest p-value according to t-test, x_3 is deleted (with p-value 0.896$> \alpha$)
- Consider t-tests from the $x_1 x_2 x_4$ model. x_4 has the largest p-value. x_4 is deleted (with p-value 0.205$> \alpha$)
- Consider t-tests from the $x_1 x_2$ model. Neither x_1 (with p-value 0$< \alpha$) nor x_2 (with p-value 0$< \alpha$) can be deleted. Conclude $x_1 x_2$ is the final model

Akaike Information Criterion

- The Akaike Information Criterion is defined as

$$AIC = n\ln(SSE) - n\ln(n) + 2p$$

- Smaller AIC is better
- If two models have same R^2 (implying same SSE), prefer the more parsimonious model

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- If two models have same R^2 (implying same SSE), prefer the more parsimonious model
- There is also Schwartz's Bayesian Criterion (BIC) and Amemiya's Prediction Criterion (APC)

AIC for cement data

Refer to cement_aic.r

Forward regression with AIC

- Start from intercept only model (AIC=71.44)
- x_4 is added to the model according to biggest drop in AIC (AIC=58.85)
- x_1 is added to the model according to biggest drop in AIC (AIC=28.74)
- x_2 is added to the model according to biggest drop in AIC (AIC=24.97)
- No more predictor can be added to the model (adding x_3 will increase AIC). Conclude $x_1 x_2 x_4$ is the final model

AIC for cement data

Refer to cement_aic.r

Backward elimination with AIC

- Start from $x_1x_2x_3x_4$ (AIC=26.94)
- x_3 is deleted to the model according to biggest drop in AIC (AIC=24.97)
- No more predictors can be deleted (deleting any predictor will increase AIC). Conclude $x_1x_2x_4$ is the final model

Stepwise regression

- In forward selection, once a predictor is added to a model, it can never be removed
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- In forward selection, once a predictor is added to a model, it can never be removed
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- Stepwise regression combines forward selection and backward elimination

Stepwise regression for cement data

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- Start from intercept only model
- x_4 is added according to p-value from t-test
- Check if any term can be dropped from x_4 model. No

Stepwise regression for cement data

Refer to cement_stepwise.r. Use $\alpha = .15$

- Start from intercept only model
- x_4 is added according to p-value from t-test
- Check if any term can be dropped from x_4 model. No
- x_1 is added according to p-value from t-test
- Check if any term can be dropped from $x_1 x_4$ model. No

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- x_1 is added according to p-value from t-test
- Check if any term can be dropped from $x_1 x_4$ model. No
- x_2 is added according to p-value from t-test
- Check if any term can be dropped from $x_1 x_2 x_4$ model.**YES**. x_4 is deleted (with p-value $0.2 > \alpha$)

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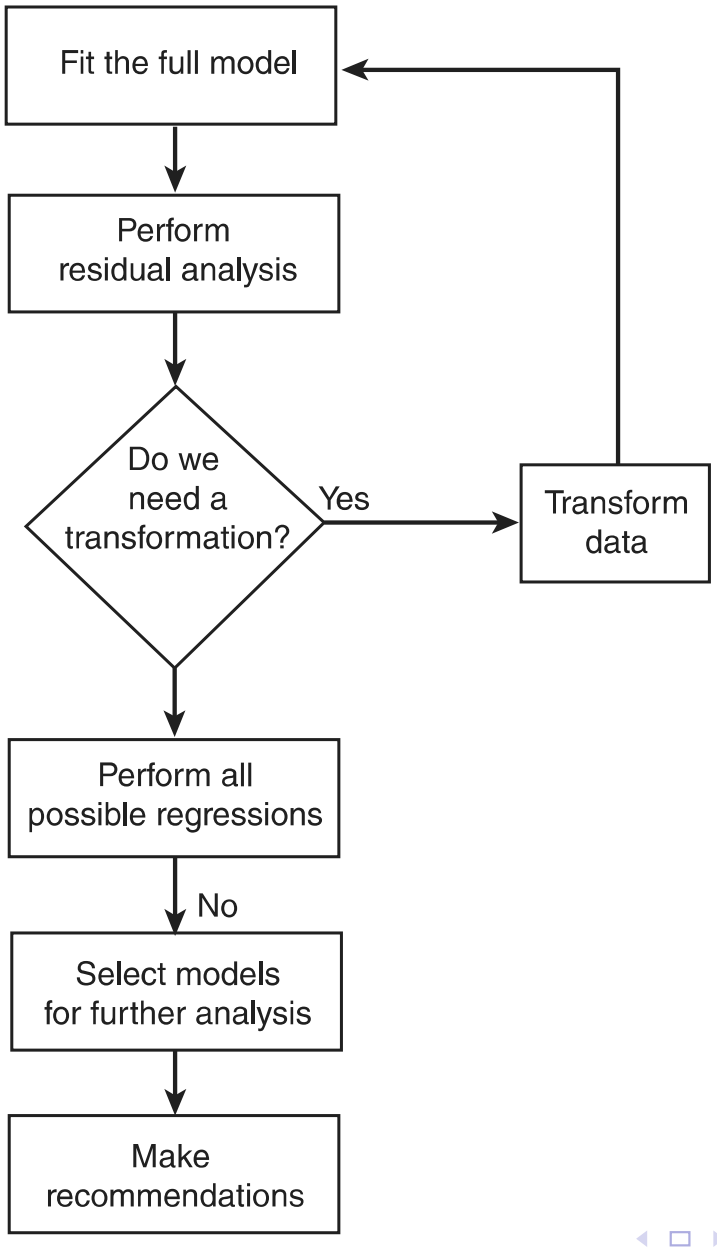
- Check if any term can be dropped from x_4 model. No
- x_1 is added according to p-value from t-test
- Check if any term can be dropped from x_1x_4 model. No
- x_2 is added according to p-value from t-test
- Check if any term can be dropped from $x_1x_2x_4$ model.**YES**. x_4 is deleted (with p-value $0.2 > \alpha$)
- Check if any term can be added to x_1x_2 model. No.
- Check if any term can be deleted from x_1x_2 model. No. Conclude x_1x_2 is the final model

Model building strategy



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- What if k is too big? If too many predictors, marginal screening (eliminate clearly irrelevant predictors) before fitting the full model